

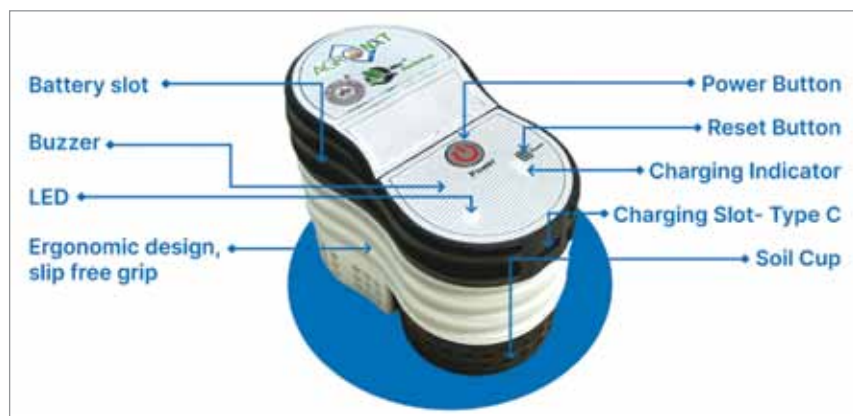
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1 TESTING SOIL INTELLIGENTLY: 100-250 SAMPLES A DAY

IIT Kanpur-incubated startup, AgroNxt, which had started out as an SaaS platform for farmers in 2016, launched a chemical-free, electronic soil-testing device, called Bhu-Parikshak, this year. It uses NIR spectroscopy and AI/ML to calculate six parameters in a soil sample instantly



Bhu-Parikshak

A typical soil testing lab in India takes as many as 7 to 30 days to test a soil sample for a farmer, according to Rajat Vardhan—the co-founder and CEO of agri-tech startup AgroNxt. But then, one probably cannot expect the 4,000-odd labs to conduct the soil tests for 140 million hectares of India’s net sown area any sooner.

This is when the smart and portable soil-testing device Bhu-Parikshak (Hindi for soil tester) comes into the picture. No technical expertise is required for its usage, but just a smartphone to connect and electricity for charging it.

Vardhan and Ashutosh Tiwari started AgroNxt as an advisory platform for the farming community. With subsequent data-driven revisions and investment from IIT (Indian Institute of Technology) Kanpur as an equity holder, the team decided to introduce a solution for one of the biggest challenges faced by the farmers: the requirement for a rapid, affordable, and chemical-free soil testing device.

Interestingly, the researchers at IIT Kanpur were already working in this direction. Soon, a team under Prof. Jayant K. Singh came out with the Bhu-Parikshak. Vardhan says,

“The institute chose AgroNxt to commercialise the product due to our agricultural and data expertise. They believed that an IoT based electronics solution, which would be plugged into our SaaS (software as a service) platform, could reach the farming community at a large scale.”

Notably, this hardware solution is not directly intended for the farmers. Vardhan points out, “More than 80% of Indian farmers are small and marginal, who’ll not be able to afford an individual device, until and unless subsidised by government.”

The company is positioning the product for the industry—the 400,000 plus agri-input shops, sugar factories, coffee and tea plantations, and FMCG (fast-moving consumer goods) companies in the agriculture supply chain, who can purchase the device and provide services to the farmers. “Microentrepreneurs, self-help groups, and farmer producer organisations could also onboard and, in turn, give soil testing as a service to the farmers by charging ₹50-100 per sample, as it has to be economically viable for these groups,” adds Vardhan.

To operate, Bhu-Parikshak needs